

SilentMethyl: Multimodal Prediction of CpG Methylation and Single-Nucleotide Variant–Associated Methylation Changes, with Calibrated Uncertainty and Independent Cross-Tissue Variant Validation

Samuel Zhang^{1,2}, Shaojun Pei^{1,2,*}, and Gil Alterovitz^{1,2,*}

¹Division of General Internal Medicine and Primary Care, Brigham and Women’s Hospital, Boston, Massachusetts, United States

²Harvard Medical School, Harvard University, Boston, Massachusetts, United States

*Authors for correspondence: spei1@bwh.harvard.edu (Shaojun Pei); gil_alterovitz@hms.harvard.edu (Gil Alterovitz)

Abstract

DNA methylation is shaped both by local sequence and by the chromatin context in which that sequence sits, but establishing which single-nucleotide variants actually change methylation remains difficult. We present SilentMethyl, a gated fusion of a pretrained DNA language model with reference epigenomic features that predicts CpG methylation and the methylation change a variant causes. Under chromosome-blocked splitting the model reaches a β -value MAE of 0.0993 and ROC-AUC of 0.968 on 26,570 held-out CpGs, outperforming six comparators including the CpGenie and DeepCpG architectures retrained on identical data. Predicted variant effects replicate across two independent cohorts — 66,495 variant–CpG pairs from an African American whole-blood meQTL study and 76,893 from tissue-matched breast — agreeing in direction and rank, surviving matching on variant–CpG distance, and pooling with no detectable heterogeneity. Two findings generalise beyond this model. Gated fusion improves absolute prediction but contributes nothing to variant-effect prediction, because context features are identical for both alleles and therefore cannot create a variant effect; and uncertainty on bounded bimodal targets is confounded by range compression, so calibration must be assessed in logit space. Predicted effect magnitude does not by itself identify clinically annotated variants.

Draft status ^{NEW}

This section is draft scaffolding. Delete it, and set `\draftmodefalse`, before submission.

New since the original draft (main.tex). Section 2.4 and Section 3.3 (independent variant evaluation on GENOA); Section 2.5 (block bootstrap and distance-matched negatives); Section 2.6 and Section 3.5 (uncertainty quantification); Section 3.4 (fusion/sequence equivalence); Section 2.7 and Section 3.6 (TCGA-BRCA tumours as a shifted target domain); Section 2.8 and Section 3.7 (ETS motif disruption). Abstract, title, introduction, discussion and limitations revised throughout.

Structural pass, 27 Aug 2026. Results cut from nine subsections to eight and re-ordered around one thesis: the model predicts methylation, its variant predictions replicate on an independent cohort, the signal runs through TF-binding disruption, the context modality cannot contribute to variant effects, and the framework yields testable hy-

potheses about individual variants. The 81 eGTEX associations were folded into Section 3.3 as the tissue-matched-but-small arm rather than deleted, because GENOA is blood and would otherwise leave no tissue-matched variant evaluation. The tumour analysis was compressed to a single paragraph and the uncertainty section from four paragraphs to two. *NCOA2* and *STK11* are retained as a thesis component, not an appendix.

Translational-scope pass, 29 Aug 2026. Added Section 2.11 and Section 3.9: two pre-registered matched-background tests of clinically annotated variant sets, both null, plus the portability and calibration arguments they bound. The pre-registrations, amendments and analysis code are in `results/journal/{clinvar,gwas}_matched_background/`. The ClinVar cohort’s effective sample size of six probes and the GWAS cohort’s power are both stated in the text rather than left implicit.

Correction pass, 29 Aug 2026. Four corrections, three of them against our own earlier text. The abstract’s motif claim asserted a specific learned regulatory relationship after Section 3.7 had already been revised to report the k -mer control that supersedes it; the abstract now matches the result. The non-draft branch of the limitations paragraph asserted repeated blocked splits, which were never run—setting `\draftmodefalse` would have printed a false methods claim at submission. Split status was added to the case studies in the abstract and in Section 3.1’s downstream subsection. Probe-QC sensitivity on `MASK_snp5_common` was added, and the two-cohort meta-analysis was promoted into the abstract.

Open question inside a written section, tested and still open. The context-only model shows no degradation at all against the tumour target and its M MAE improves (Section 3.6). Genuine robustness of coarse chromatin signal and bounded-range compression from a less bimodal tumour β distribution both predict this. The compression limb was tested directly on the `MASK_snp5_common` strata, where the same signature appears, by resampling unflagged probes to the flagged stratum’s β histogram (`scripts/29_range_compression_check.py`). It did not resolve: the differences being attributed are 0.005–0.010 in ROC-AUC and 0.0006–0.012 in beta MAE, against a combined standard error of roughly 0.007, and the three models do not behave alike. The test is underpowered relative to the effect it was asked to explain, the result is recorded in `results/journal/range_compression/`, and neither limb may be asserted in the text.

Not yet written, in priority order. (i) repeated chromosome-blocked splits, four additional folds at seed 42—the only remaining model training; (ii) conformational prediction intervals; (iii) HOCO-MOCO as a second motif library.

Completed since the previous pass. The head-to-head against CpGenie and DeepCpG is done on both tasks. Absolute prediction: Section 2.10 and Table 1, both models reimplemented from released source, trained on our splits, tuned over their authors’ own grids, three seeds each. Variant effects: both scored on the identical GENOA and eGTEx pairs and compared to fusion on a shared distance-matched cohort with paired intervals (`scripts/30_paired_model_comparison.py`, Section 3.3). The variant-effect arm returned eleven nulls in twelve comparisons and is written up as convergence across architectures; it was run after the absolute-prediction arm and reversed the ordering we had provisionally read off unpaired per-model estimates.

Run, and uninformative for a structural reason—recorded so its absence from the claims is not read as a suppressed result. The ClinVar matched-background test was pre-registered in `preregistration.json` before any score existed: primary cohort the 35 held-out non-truncating variants, primary statistic the count with matched-background tail probability below 0.05 against a binomial null of 0.05, two-sided, with a power statement declared in advance. It was subsequently built against a synthetic comparator pool drawn from the general held-out population and run. The result is reported in Section 3.9. It carries no weight because the 35 variants span only six distinct probes, 25 of them on a single CpG: the apparent depletion at variant level disappears once the interval is clustered on probe, leaving an effective n of six that no comparator pool can rescue. An earlier version of this section stated the test had been abandoned and never run; that was true when written and is corrected here.

Bibliography. New citation keys required in `references.bib`: `shang2023genoa`, `avsec2025alphagenome`, `ying2024methylogpt`, `gemma2012`, `castrocolin2013jaspar`, `zeng2017cpgenie`, `angermueller2017deepcpg`, `buniello2019gwascatalog`. If `references.bib` is absent at build time, this document falls back to a key-only reference list so that it still compiles.

1 Introduction

DNA methylation is a widely studied epigenetic modification involved in gene regulation, associated with genomic context, chromatin organization, and transcription [1, 2], and frequently dysregulated in disease. Single-nucleotide variants (SNVs) can influence local epigenetic states, but validating their direct regulatory effects is difficult because matched genetic and epigenetic data are limited. This motivates modeling DNA methylation to identify potentially functional variants.

Existing deep learning methods predict DNA methylation and related regulatory measures from local DNA sequence, neighboring methylation, and long-range sequence context [3–5]. While DNA sequence contains motifs and local regulatory patterns, sequence alone misses the epigenetic context in which they operate. Reference epigenomic data, including chromatin accessibility, histone marks, and evolutionary conservation, can supply part of this missing information. Pre-

dicting methylation at a locus and estimating the effect of a base change on methylation are also distinct tasks: a model that predicts baseline methylation accurately may still estimate variant-induced changes poorly.

Several frameworks for evaluating paired sequence changes already exist. CpGenie predicts CpG methylation and validates REF–ALT sequence changes against mQTLs [6]. INTERACT focuses on brain methylation, DeepMethylation/DDM combines sequence with reference tissue data, and Methven combines sequence with ATAC-seq accessibility [7–9].

^{REV}A specific gap motivates this work. Recent long-context variant-effect models—Enformer, Borzoi and AlphaGenome—predict many regulatory modalities at megabase scale but do not predict CpG methylation [5, 10]. Conversely, methylation foundation models such as MethylGPT operate on measured methylation profiles rather than sequence, and so cannot score an unobserved variant at all [11]. Sequence-based prediction of *variant effects on methylation* therefore sits between two active literatures without being served by either.

Existing approaches also share two methodological limitations. Most quantify the contribution of sequence and epigenomic inputs only through aggregate model-performance differences, and most do not account for reverse-complement equivariance [12]. SilentMethyl addresses both: gated fusion exposes the relative weighting of sequence and context at each individual probe, and reverse-complement stability is enforced by random orientation during training and by averaging forward and reverse-complement predictions at inference, which makes the reported output exactly RC-invariant by construction.

SilentMethyl uses synonymous mutations as a focused baseline for variant prioritization, since they change a base without changing the encoded amino acid while still potentially affecting splice recognition, RNA structure, transcript stability, translation, and exonic regulatory sequence [13, 14]. Several studies already link synonymous variants to DNA regulatory features: a synonymous *MLH1* c.27G>A allele was found in individuals with constitutional *MLH1* methylation [15]; synonymous *TP53* variants altering CpG content are linked to intragenic demethylation and alternative promoter usage [16]; synonymous *ADAMTS13* variants alter molecular properties by creating and destroying CpG sites [17].

^{REV}This work makes four contributions. (1) A gated multimodal architecture that improves CpG methylation prediction over either modality alone and exposes per-locus modality weighting. (2) Independent variant evaluation at a scale that supports stratified inference: 66,495 held-out variant–CpG pairs from an African American whole-blood cohort, evaluated against a distance-matched negative control with intervals that respect linkage disequilibrium. (3) A demonstration that uncertainty evaluation on bounded bimodal targets is confounded by range compression, with a scale-stability diagnostic that distinguishes genuine uncertainty signal from metric artifact. (4) A negative result with a mechanistic explanation: gated fusion cannot contribute to variant-effect predic-

tion while its context features are allele-invariant. The last two apply to any sequence-plus-context model performing variant scoring.

2 Methods

2.1 Data construction

Methylation beta labels were constructed from the TCGA-BRCA HM450 matrix [18, 19]. Using a sample-type-11 filter, 97 solid-tissue-normal columns were selected from 893 total, verified to represent 97 unique participants without duplicate samples.

Single-probe methylation was represented as the median of patient beta values. For CpG i , if O_i is the set of patients with a recorded measurement at CpG i , then

$$\begin{aligned}\beta_i &= \text{median}_{s \in O_i} \beta_{i,s}, & y_i &= \mathbf{1}[\beta_i > 0.5], \\ \tilde{\beta}_i &= \max(10^{-4}, \min(1 - 10^{-4}, \beta_i)), \\ M_i &= \log_2 \left(\frac{\tilde{\beta}_i}{1 - \tilde{\beta}_i} \right).\end{aligned}$$

The beta-to- M transformation follows [20].

All probes required an exact CpG-centered DNA sequence and a valid reference context window. None of the 418,486 probes in the combined dataset were flagged by the HM450 MASK_general filter. Median probe coverage was 97 of 97, and 95.1% of loci had at least 78 valid patients. Probe coordinates and sequence were taken from the hg38 HM450 annotation resource [21, 22].

Data were split by chromosome before training: chromosomes 10–11 for validation, chromosomes 8–9 for test, remaining chromosomes for training (Table S1). Splitting by chromosome ensures no sequence windows overlap between partitions.

Each DNA input was a 1,000-bp CpG-centered sequence with the target CpG nucleotides at zero-indexed positions 499 and 500. The epigenetic input comprised seven MCF-10A attributes—ATAC-seq accessibility and six histone marks (H3K4me3, H3K27ac, H3K27me3, H3K9me3, H3K36me3, H3K4me1)—averaged across a CpG-centered 100-bp window, plus two PhyloP scores for the two nucleotides of the central CpG. A missingness mask accompanied each feature; missing values were imputed with the feature’s global median. Reference data were derived from shared resources and were not patient-specific.

2.2 Models, training, and evaluation

Figure 1 summarises target construction, the chromosome partition, the gated architecture, and the paired scoring and evaluation paths described in this section. The sequence module comprised a pretrained DNABERT-2 (117M) transformer [23] followed by a convolutional layer and masked attention pool-

ing:

$$\alpha_t = \frac{\exp(s_t)}{\sum_{u \in V} \exp(s_u)}, \quad \mathbf{h}_{\text{seq}} = \sum_{t \in V} \alpha_t \mathbf{e}_t,$$

where V contains non-padding positions and $\mathbf{e}_t$ is the embedding at position t , producing a 768-dimensional $\mathbf{h}_{\text{seq}}$.

The context module was an MLP mapping the 18-dimensional epigenetic input (9 features + 9 missingness masks) to a 768-dimensional $\mathbf{h}_{\text{ctx}}$.

After LayerNorm produced $\tilde{\mathbf{h}}_{\text{seq}}$ and $\tilde{\mathbf{h}}_{\text{ctx}}$, the two were concatenated and passed through a gating MLP yielding scalars g_{seq} and g_{ctx} . The fused embedding was

$$\mathbf{z} = g_{\text{seq}} \tilde{\mathbf{h}}_{\text{seq}} + g_{\text{ctx}} \tilde{\mathbf{h}}_{\text{ctx}},$$

fed to two prediction heads. The regression head predicted M , converted by $\hat{\beta} = 2^{\hat{M}} / (1 + 2^{\hat{M}})$; the classification head predicted $\hat{y} = \mathbf{1}[\sigma(\text{logit}) > 0.5]$. The objective was Huber loss ($\delta = 1.345$) on regression plus binary cross-entropy on classification.

Sequence and context models were trained independently with AdamW, with DNABERT-2 fully fine-tuned. Sequence, context, and fusion training used up to 10, 15, and 10 epochs at learning rates 2×10^{-5} , 10^{-3} , and 5×10^{-4} . During fusion training the component encoders were frozen and only normalization, gating, and prediction modules were updated.

During training, forward or reverse-complement orientation was chosen with equal probability per probe. For seeds 42, 43, and 44 the checkpoint was selected by lowest validation beta MAE. Primary metrics were RMSE and MAE for β and M , and ROC-AUC for classification.

2.3 Paired sequence scoring and sSNV construction

For each SNV the predicted methylation difference was averaged across strands:

$$\Delta\hat{\beta} = \frac{(\hat{\beta}_{\text{MUT,FWD}} - \hat{\beta}_{\text{WT,FWD}}) + (\hat{\beta}_{\text{MUT,RC}} - \hat{\beta}_{\text{WT,RC}})}{2},$$

with $\Delta\hat{M}$ defined analogously. Because the reported prediction is $\frac{1}{2}[f(x) + f(\text{RC}(x))]$, it is exactly invariant under reverse complementation.

Tagged somatic sSNVs were obtained through the GDC with amino-acid preservation verified against hg38 and GENCODE v44 [19, 24]. Variants were matched to the nearest HM450 probe; 440 unique variants remained after removing MASK_general-flagged probes and variants outside the 1,000-bp window. Variants were prioritized by $|\Delta\hat{\beta}|$ from the three-seed fusion ensemble.

2.4 Independent variant evaluation: GENOA ^{NEW}

External validation used cis-meQTL summary statistics from GENOA, an African American whole-blood cohort [25].

Figure workflow.pdf not present in this build.

Figure 1: SilentMethyl data construction, model architecture, and evaluation workflow. Panel A shows methylation target construction, chromosome partitioning, and model inputs. Panel B shows FWD-RC orientation and staged training. Panel C shows the gated multimodal architecture. Panel D shows paired WT-MUT scoring, held-out probe testing, external mQTL validation, and synonymous-variant ranking.

Three properties make it suitable: it is a different tissue, a different ancestry, and a different array generation from the training data, and it is openly released.

Harmonisation. Summary statistics are reported in hg19. The CpG was never lifted over: array probe identifiers are platform-stable, so the reported probe joins directly to the hg38 manifest and inherits hg38 coordinates. Only SNP positions required hg19→hg38 conversion. Of 942,186 candidate pairs, 731 (0.1%) had neither reported allele matching the hg38 reference base and were discarded; this rate is itself evidence that the liftover and strand convention are correct.

Effect-allele alignment. GENOA effect sizes are GEMMA output, in which β is the effect per copy of allele1, the minor allele [26]. The model’s prediction is the effect of substituting ALT for the hg38 reference. These directions coincide only when ALT is the minor allele. In 106,628 pairs (11.3%) the hg38 reference base is the minor allele, so ALT is the major allele and the two conventions are opposed. All comparisons therefore use the published effect re-signed to the REF→ALT direction. This correction is silent if omitted—it does not produce an error—and it attenuates both signed correlation and direction agreement.

Strata. Pairs were labelled by the split membership of their target probe, read from the actual training partitions rather than inferred from chromosome number. Only the held-out stratum supports an independent-validation claim: 66,495 pairs, 39,657 unique variants, 19,081 of the 26,570 held-out probes. Pairs whose variant falls at the target CpG itself were excluded (the architecture is defined around a CpG at window indices 499:501, and such array measurements are a known SNP-under-probe artifact), as were pairs outside the 1,000-bp window, for which the model returns a delta of exactly zero. Variants that create or destroy a CpG dinucleotide have a near-deterministic direction requiring no model and are reported as

a separate stratum throughout; unless stated otherwise, results below refer to the 42,866 non-CpG-altering held-out pairs.

2.5 Statistical framework ^{NEW}

Dependence. The held-out pairs comprise 39,657 variants in linkage disequilibrium across 19,081 probes and are far from independent. All intervals are percentile bootstrap intervals resampling whole 1-Mb genomic blocks rather than individual pairs; row-level resampling would treat correlated observations as independent and yield intervals several times too narrow.

Distance matching. Significant meQTLs lie closer to their target CpG than tested-but-null pairs (median 191 versus 258 bp), and predicted effect magnitude increases with proximity ($\rho = -0.29$ between $|\Delta M|$ and distance). Variant-CpG distance alone therefore separates the two classes, and an unmatched discrimination estimate partly measures distance rather than regulatory effect. Each significant pair was matched to a null pair at the same base-pair distance, sampled without replacement so that no null could be reused. Within the matched cohort, distance alone achieves AUROC 0.500 by construction, so residual discrimination is attributable to the model. The distance-only baseline is reported alongside every discrimination estimate.

Paired comparison between models. ^{NEW} Two models evaluated separately receive independent draws of matched negatives, so part of any gap between their reported AUROCs is matching noise rather than model difference. Marginal intervals are also the wrong instrument for a between-model claim: estimates computed on the same variants are strongly correlated, so overlapping marginal intervals do not imply that a difference is absent. Every between-model comparison in this work therefore uses a single shared distance-matched cohort — matching depends only on the significance label and the variant-CpG distance, both properties of the variant rather

than of the model — with the difference bootstrapped over the same 1-Mb blocks for both models inside each resample. Model-specific point estimates on the shared cohort differ slightly from the per-model values in Table 2 for exactly this reason; the differences, not the marginal values, are the comparison.

2.6 Uncertainty quantification ^{NEW}

Averaging forward and reverse-complement predictions makes the output invariant but discards the disagreement between them. That disagreement is available at every locus at no additional cost and was evaluated as a single-model uncertainty estimate against a three-seed ensemble standard deviation, a zero-parameter boundary heuristic $-\lvert\hat{\beta} - 0.5\rvert$, and a random control, using risk–coverage curves and area under the risk–coverage curve.

Because β is bounded on $[0, 1]$ and bimodal, absolute β error is mechanically compressed near the boundaries. Every comparison was therefore repeated on the unbounded M scale, and additionally within strata of predicted β at 10, 20, and 50 levels, so that an estimator’s apparent skill could be separated from headroom in the target.

2.7 Tumour cohort as a shifted target domain ^{NEW}

The TCGA-BRCA matrix already used for training contains 791 primary-tumour columns that were never used. Of these, 699 come from participants absent from the 97 solid-tissue-normal samples used to build the training targets; those 699 form the primary cohort, and the full 791 are reported as a supplement because the remaining 91 participants make that comparison paired rather than independent.

One property of this design must be stated plainly. Model inputs are the reference sequence and MCF-10A context features, neither of which depends on disease state, so *predictions are identical for normal and tumour*. This evaluation holds predictions fixed and swaps the target. It therefore measures the component of the tumour methylome determined by sequence and reference chromatin context, and where the model degrades is where methylation has been reprogrammed by something the model cannot observe. It is not, and is not presented as, generalisation to a new input domain.

2.8 Motif disruption analysis ^{NEW}

JASPAR CORE vertebrates [27] (877 matrices, of which 831 met a 200-variant coverage threshold) was scanned across the 42,866 non-CpG-altering held-out GENOA pairs. Counts were converted to log-odds matrices against a uniform background with a Laplace pseudocount, and scores were expressed relative to each matrix’s own range so that a threshold means the same thing across motifs of different length and information content; 0.80 is the conventional cutoff.

Both strands were scanned, since motifs are strand-specific.

For each pair, the best-scoring wild-type occurrence covering the variant was located, and the mutant was scored *at that same offset and strand*: choosing the optimum separately per allele would allow apparent disruption to be the motif relocating to a different site. Per-factor coupling is the Spearman correlation between the change in relative motif score and the predicted change in M , across pairs covered by that motif, with Benjamini–Hochberg correction across factors.

Three controls were applied, each targeting a distinct way the result could be an artifact. A *null distribution* across all tested motifs distinguishes a factor-specific effect from a global offset in which disrupting any motif shifts predictions one way. *Partial correlations* residualise local GC content and variant–CpG distance, since motifs concentrate in GC-rich sequence and GC-rich sequence sits nearer CpG islands. A further partial adds *canonical substitution class* (six categories, pyrimidine reference), because compositionally biased motifs imply biased base changes: weakening a purine-rich motif usually means a G/A→C/T substitution, and pyrimidine-rich sequence is more methylated on average.

Because motif families share a core, top-ranked factors are not independent observations. Pairwise Jaccard overlap of the covered-variant sets is reported so that redundancy is measured rather than assumed.

A binary inside-versus-outside-a-motif contrast is not usable here: with the full library scanned, every variant falls inside some occurrence at the 0.80 cutoff, leaving no background. Contrasts therefore use top-versus-bottom quartiles of disruption magnitude, matched on exact variant–CpG distance and GC quintile.

2.9 External eGTEx mQTL analysis

Significant eGTEx Breast Mammary Tissue lead mQTLs ($q < 0.05$) were overlapped with test-set CpGs [28]. After excluding variants outside the sequence window or altering their target CpG, 81 associations with 70 unique variants remained. Predicted ALT–REF changes were compared with reported allele slopes using signed and absolute Spearman correlation, sign agreement, and AUROC. A sensitivity analysis dropped mutations within or immediately adjacent to a probe. A matched analysis tested whether predicted magnitude separated significant leads from high- q associations ($q \geq 0.5$) matched on chromosome and minor allele frequency (35 pairs from 53 significant and 51 high- q variants).

2.10 Published methylation architectures, reimplemented ^{NEW}

Comparison against CpGenie [6] and DeepCpG [3] required a decision about how to run them. Both are 2017-era Keras/Theano/TensorFlow-1 codebases that do not install on current hardware, and the published CpGenie weights were trained on GM12878 lymphoblastoid cells rather than breast tissue, so scoring those weights against our targets would impose a cross-tissue handicap and produce a comparison

favourable to us for the wrong reason. We therefore reimplemented both architectures and trained them on our splits.

Layer specifications were taken from the released source rather than from the papers. CpGenie follows `cnn/seq_128x3_5_5_2f_simple.template`: three convolution/max-pool stacks (128, 256 and 512 filters of width 5, pooling width 5 stride 3) into two 64-unit dense layers with dropout, max-norm 3 on the convolutional weights, and RMSprop. DeepCpG follows `CnnL2h128 in deepcpG/models/dna.py`: convolution 128@11, max-pool 4, convolution 256@3, max-pool 2, dense 128, dropout. Published DeepCpG additionally carries a CpG module, which reads observed methylation at neighbouring CpGs across cells, and a joint module. We use the DNA module alone, for two reasons. The neighbour input does not exist in a bulk cohort-median design, and more fundamentally it is measured rather than predicted, so it is identical for the reference and alternate alleles and contributes exactly zero to any predicted variant effect — the same allele-invariance that governs our own context tower (Section 3.4). The DNA module is therefore not a weakened DeepCpG but the only part of it that can address this task. As a check on our reading of the layer specification, the DeepCpG dense layer computes to 3,997,824 parameters at a 1,000 bp input against the 4,100,000 its own docstring reports.

Both were trained on the identical training split with the identical reverse-complement augmentation and the identical loss used for our models, and evaluated with the same reverse-complement averaging. Hyperparameters were selected on the validation split over the grids the authors themselves declare: for CpGenie, dropout $\in \{0.3, 0.5, 0.7\}$ crossed with learning rate $\in \{0.01, 0.001, 0.0001\}$, the hyperas search in their template; for DeepCpG, dropout $\in \{0.0, 0.3, 0.5\}$ at their default learning rate. The test split was never used for selection. Selected settings were dropout 0.3 and learning rate 0.001 for CpGenie, and dropout 0.0 for DeepCpG; both sat at the lower edge of the dropout range offered, which we report because it means a lower dropout than either grid contains might serve these baselines slightly better.

Two deviations are deliberate. Both originals predict a binary methylation state, so we attached our dual head—M-value regression plus a binary logit—to each published trunk, which scores every model on identical targets and isolates the architecture rather than the output parameterisation. And both were trained at our 1,000 bp window rather than their own defaults, so the input is identical across the comparison.

2.11 Matched-background tests of clinically annotated variant sets ^{NEW}

Two pre-registered tests ask whether predicted effect magnitude is larger for clinically annotated variants than for comparable variants. Both compare each variant against a background matched on substitution class (SBS96), CpG effect, and distance to the target CpG, using the tiered matching of the candidate analysis; the reported statistic is the variant’s percentile within its own matched background, and the two-

sided empirical tail probability.

Background construction. Sequence and epigenomic features in this project are properties of the probe rather than of the variant, so a comparator is a different single-base substitution inside a held-out probe window. We generated 14,996 such variants across 5,000 held-out probes, positions and alternate alleles drawn uniformly at random within the 1,000 bp model window excluding the centred CpG, capped at three per probe (fixed seed). A generated background is disjoint from the tested cohorts by construction, carries no allele-frequency confound, and can be made large enough that matching resolves at strict tiers.

Cohorts. The first cohort is the 35 held-out non-truncating ClinVar-pathogenic variants from the literature screen. The second is breast-cancer associations from the GWAS Catalog on the held-out chromosomes within 500 bp of a held-out probe; reference alleles were taken from Ensembl (GRCh38) and alternate alleles from the catalogue’s risk allele, giving 32 variants on 32 distinct probes. The coordinate convention was validated by requiring the reference base read from the probe window to match the Ensembl reference allele, which held for 32 of 32 at the adopted offset against 21.9% and 37.5% at the alternatives.

Statistics and pre-registration. Cohort definitions, both statistics, the clustering unit and the power calculation were fixed in writing before any variant was scored, together with the interpretation rule for each possible outcome; amendments are timestamped and recorded. Because variants at the same CpG share the window and the context features, *the probe is the independent unit* and all interval estimates cluster on it. A context check comparing cohort and background on all nine epigenomic features is run before any result is interpreted.

3 Results

3.1 Multimodal fusion outperforms single-modality models

On the 26,570-probe test set the fusion model was strongest across all metrics and seeds. The context MLP established a baseline beta MAE of 0.1395 ± 0.0001 and ROC-AUC 0.9187 ± 0.0001 . The sequence model reduced beta MAE by 21.2% (0.1099 ± 0.0008) and raised ROC-AUC to 0.9569 ± 0.0007 . Fusion improved beta MAE by a further 9.6% (0.0993 ± 0.0025) and raised ROC-AUC to 0.9680 ± 0.0021 (Fig. 2). Predictions were stable at lower coverage: on probes measured in at least 78 of 97 patients the seed-42 fusion model reached beta MAE 0.09669 and ROC-AUC 0.97066.

Improvement was consistent across seeds and across genomic regions. Fusion improved sequence-only MAE by 0.01043 (95% CI 0.00951–0.01132) and ROC-AUC by 0.01053 (0.00909–0.01207) under a 257-block resampling design.

The gain is not a restatement of sequence composition. ^{NEW}
Two classical baselines were fitted on the identical splits: a

Table 1: Held-out chr8–9 performance against classical baselines and published methylation architectures.^{REV} Mean $\pm$ SD across three seeds on 26,570 CpGs. Every model is fitted on the identical training split with hyperparameters selected on the identical validation split, so no tissue or data advantage separates them. The ridge baselines have their regularisation strength tuned; CpGenie and DeepCpG were reimplemented from their published source and tuned over the authors’ own declared grids (Section 2.10). Parameter counts: CpGenie 2.0 M, DeepCpG 4.1 M, DNABERT-2 encoder 117 M.

Metric	Composition (3 feat.)	<i>k</i> -mer ridge (2,772 feat.)	Context (9 feat.)	CpGenie (reimpl.)	DeepCpG (DNA module)	Sequence DNABERT-2	Fusion full
M-value MAE	1.9600	1.6866	1.4574 $\pm$ 0.0023	1.3858 $\pm$ 0.0084	1.3412 $\pm$ 0.0092	1.1941 $\pm$ 0.0070	1.0971 $\pm$ 0.0192
Beta MAE	0.1954	0.1565	0.1395 $\pm$ 0.0001	0.1281 $\pm$ 0.0011	0.1253 $\pm$ 0.0004	0.1099 $\pm$ 0.0008	0.0993 $\pm$ 0.0025
ROC-AUC	0.8748	0.9190	0.9187 $\pm$ 0.0001	0.9406 $\pm$ 0.0005	0.9437 $\pm$ 0.0009	0.9569 $\pm$ 0.0007	0.9680 $\pm$ 0.0021

three-feature composition model (GC fraction, CpG count, CpG observed/expected) and a ridge regression over 2,772 reverse-complement-collapsed *k*-mer counts, $k = 1 \dots 6$. The *k*-mer model is a strong comparator — it reaches ROC-AUC 0.9190, statistically indistinguishable from the nine-feature chromatin model at 0.9187, so 2,772 sequence features carry about as much information about methylation level as measured chromatin state does. The sequence tower alone nevertheless reduces its beta MAE by 29.8% (0.1565 to 0.1099) and fusion by 36.6% (to 0.0993). Pretrained representation is therefore doing work that *k*-mer composition cannot, and the claim is measured on identical data rather than asserted.

Held-out probes are not near-duplicates of training probes.^{NEW} Chromosome-blocked splits prevent positional leakage but not sequence-similarity leakage: segmental duplications and paralogous promoters can place near-identical windows on different chromosomes. Estimating Jaccard similarity over canonical 31-mers (MinHash, sketch 128) between every held-out probe window and its nearest training window, the median is 0.000 and the 99th percentile 0.094; four of 26,570 test probes (0.02%) exceed 0.5. Held-out performance is not memorisation of near-duplicate sequence.

The held-out chromosomes are also modestly more methylated than the training chromosomes (median beta 0.642 versus 0.550; 46.5% versus 42.3% of probes above 0.7), with comparable spread. A model that had learned the training distribution would be biased low under that shift; mean signed beta error is -0.002 to -0.004 across seeds, so predictions track the shift rather than regressing toward the training mean.

Probe quality control does not account for the result either. The HM450 manifest flags 15.4% of probes as MASK_snp5_common—a common SNP within 5 bp of the interrogated CpG, where the measured beta may be partly a genotype artefact—covering 3,105 of the 26,570 held-out probes (11.7%). Restricting to the 23,465 retained probes leaves fusion beta MAE unchanged at 0.0993 and raises ROC-AUC from 0.9680 to 0.9689. The flagged probes are the harder stratum for all three architectures (fusion 0.9530, sequence-only 0.9362, context-only 0.9096), which locates the deficit in the probes rather than in any model exploiting genotype signal. No model was re-run; predictions written at test time were re-scored and reproduced the published values to within 5×10^{-4} .

Figure model_incremental_performance.png
not present in this build.

Figure 2: Held-out chromosome performance. Gray lines connect the same training seed. (A) M-value MAE. (B) beta MAE. (C) ROC-AUC.

Against the published architectures the ordering is monotone and the margin is substantial. CpGenie and DeepCpG, trained on our splits with hyperparameters selected over the grids their own authors declared, reach beta MAE 0.1281 ± 0.0011 and 0.1253 ± 0.0004 with ROC-AUC 0.9406 and 0.9437. Both clearly exceed the classical baselines—CpGenie improves on *k*-mer ridge by 18.1%—confirming that a learned convolutional encoder extracts more from the same sequence than fixed *k*-mer counts. Both are in turn exceeded by the pretrained encoder: sequence-only DNABERT-2 lowers beta MAE 12.3% below the better of them, and the full fusion model lowers it **20.8%**, with M-value MAE 18.2% lower and ROC-AUC 0.0243 higher.

Two things temper that comparison and we state them rather than leaving them implicit. The DNABERT-2 encoder carries roughly 117 M parameters against CpGenie’s 2.0 M and DeepCpG’s 4.1 M, and it arrives pretrained on genomic sequence, so the gain reflects capacity and pretraining together rather than architecture alone. And both published models were designed for a binary methylation-state objective; we attached our dual regression-plus-classification head to their published trunks so that all seven columns are scored on identical targets, which is the fairer comparison but is not the configuration their authors evaluated.

3.2 Reference context contributes across regulatory conditions

Fusion reduced beta MAE in every stratum of GENCODE region, CpG island status, ATAC-seq quartile, and H3K27ac quartile. By GENCODE annotation the reduction ranged from 0.0088 at intergenic probes to 0.0130 at UTRs. By island status the largest gain was at shores (0.0162). Fusion lowered beta MAE by 0.0139 in the highest ATAC-seq quartile and 0.0147 in the highest H3K27ac quartile (Fig. 3). Context therefore contributes most where regulatory state is expected

Figure fusion_gain_by_epigenomic_context.png
not present in this build.

Figure 3: Fusion gain across epigenomic contexts. Intervals re-sample 1-Mb regions within each stratum. (A) CpG island status. (B) ATAC-seq quartile. (C) H3K27ac quartile.

to carry additional information.

3.3 Independent variant evaluation ^{REV}

Variant-effect predictions were tested against two external cohorts chosen to differ from the training data along different axes. The predictions validate in both, and the pattern of where they are strong and where they attenuate is the pattern the architecture predicts rather than an unexplained scatter of results.

In matched tissue, agreement is strong. The complete eGTEx Breast Mammary association set (2.75 billion tests) yields 76,893 held-out variant-CpG pairs. Significance follows the study’s own two-stage permutation procedure rather than a genome-wide constant: probe-level FDR ≤ 0.05 gives a calibrated nominal cutoff of $p \leq 1.483 \times 10^{-5}$, declared before any eGTEx score existed. On the resulting 418 significant non-CpG-altering pairs the three-seed fusion ensemble reaches signed $\rho = 0.246$ (0.114–0.402) and 59.6% direction agreement (0.547–0.652), with a calibration slope of 1.60 (1.13–2.28). Every metric exceeds the cross-tissue cohort below.

The 81 previously published eGTEx lead mQTLs, a subset of this set, give $\rho = 0.610$ (0.452–0.731, $p \approx 1 \times 10^{-9}$) and 79.0% direction agreement. Lead variants are enriched for large, precisely measured effects, so that figure is not comparable to a whole-cohort estimate and is reported as the lead-variant stratum rather than as the headline.

Across tissue and ancestry, the effect replicates at scale but attenuates. Scoring 66,495 held-out variant-CpG pairs raised the evaluation to 39,657 unique variants across 19,081 of the 26,570 held-out probes, at a scale that supports stratified inference rather than a single pooled estimate (Table 2). Because GENOA measured whole blood while the model’s context features are breast, this is cross-tissue transfer; the sequence-only model, which uses no tissue-specific input, is reported alongside as the tissue-agnostic comparator.

Table 2: Independent variant evaluation on GENOA. Held-out stratum, non-CpG-altering pairs, three-seed ensemble. Brackets are 95% intervals from a block bootstrap over 1-Mb blocks. Signed ρ and direction agreement are computed on genome-wide significant pairs ($p < 5 \times 10^{-8}$, $n = 4,037$); AUROC uses all 42,866 pairs.

Metric	Fusion	Sequence
Signed ρ	+0.152 [+0.117, +0.188]	+0.150 [+0.109, +0.184]
Direction agreement	0.553 [0.536, 0.570]	0.554 [0.538, 0.573]
AUROC, distance-matched	0.570 [0.554, 0.585]	0.562 [0.546, 0.578]
AUROC, unmatched	0.600 [0.586, 0.614]	0.603 [0.589, 0.617]
<i>AUROC, distance only</i>	0.595 [0.583, 0.609]	

Table 3: Variant-effect performance in two external cohorts. Held-out stratum, significant non-CpG-altering pairs, three-seed fusion ensemble. Each cohort uses its own pre-declared significance definition. Brackets are 95% block-bootstrap intervals.

	eGTEx breast tissue-matched	GENOA blood cross-tissue, cross-ancestry
<i>n</i>	418	4,037
Signed ρ	+0.246 [+0.114, +0.402]	+0.152 [+0.118, +0.186]
Direction agreement	0.596 [0.547, 0.652]	0.553 [0.536, 0.571]
Calibration slope	+1.60 [+1.13, +2.28]	+1.17 [+0.96, +1.44]
<i>Meta-analysis: $\rho = 0.178$ [0.068, 0.283], $p = 1.6 \times 10^{-3}$, $I^2 = 0\%$</i>		

The two cohorts agree, and combining them is the result. Compared like with like — all significant non-CpG-altering pairs in each — the tissue-matched cohort exceeds the cross-tissue one on every metric (Table 3). An inverse-variance meta-analysis on Fisher-transformed ρ , weighted by independent 1-Mb linkage blocks rather than pair counts, gives $\rho = 0.178$ (0.068–0.283, $p = 1.6 \times 10^{-3}$) with **Cochran’s** $Q = 0.57$ ($p = 0.45$), $I^2 = 0\%$. The cohorts differ in tissue, genetic ancestry, donor count, platform generation and genotyping pipeline, and show no detectable heterogeneity. The effect is not a property of either cohort’s idiosyncrasies.

The relationship is quantitative, not merely ordinal. Regressing reported effect on predicted ΔM over the significant stratum gives a slope of +1.17 (+0.96, +1.44) in GENOA and +1.60 (+1.13, +2.28) in eGTEx; the sequence-only model gives +0.98 (+0.81, +1.20) and +1.25 (+0.82, +1.84). Rank statistics establish that the ordering is right; a slope whose interval excludes zero establishes that predicted magnitude scales with measured effect. The numeric value should not be read as calibration — reported effects are on an inverse-normal-transformed scale and predictions are in M -value units — and ordinary least squares is attenuated by error in the predictor, so the underlying relationship is at least this steep.

An internal negative control. GENOA publishes every cis pair tested rather than only the meQTLs discovered, and 71% of held-out non-CpG-altering pairs have $p > 0.05$. Stratifying by association strength shows agreement rising with the strength of the underlying association and reaching chance where no association exists (Table S2, Fig. 4). The null stratum is statistically indistinguishable from zero on both metrics, providing a negative control internal to the same dataset. Intermediate strata are not strictly monotone and their intervals

Figure significance_gradient.png
not present in this build.

Figure 4: Agreement with GENOA rises with association strength and reaches chance where associations are null. Signed Spearman correlation (left) and direction agreement (right) across seven strata of reported p -value, with 95% block-bootstrap intervals.

overlap; the interpretable contrast is significant versus null.

A second, independent negative control: the signal is spatially localised. Agreement is highest within 100 bp (0.605, 0.559–0.654 in the 50–100 bp bin), declines through the intermediate bins, and reaches chance at the edge of the model’s receptive field (0.497, 0.459–0.535 at 400–501 bp; Table S3). A confound is unlikely to produce this: linkage structure, GC content and probe properties have no reason to decay with distance from the CpG, whereas a 1,000 bp window centred on it must. With the association-strength null, the analysis rests on two orthogonal negative controls, one statistical and one positional.

Discrimination is not explained by distance alone. Variant–CpG distance by itself separates true meQTLs from null pairs at AUROC 0.595 (0.583–0.609), statistically indistinguishable from the model’s unmatched 0.600 (0.586–0.614). We state this explicitly: *marginally, predicted effect magnitude does not outperform distance*. After matching each significant pair to a null pair at identical distance, however, discrimination remains at 0.570 (0.554–0.585), and exceeds chance in every distance bin examined. The model’s signal is therefore largely orthogonal to distance rather than a restatement of it (Fig. 5).

Discrimination is what separates the model from a composition baseline. ^{NEW} Scored through the identical evaluator on the identical pairs, the k -mer ridge reaches signed $\rho = 0.100$ (0.059–0.133) — so it carries some compositional signal — but its distance-matched AUROC is 0.503 (0.488–0.518), indistinguishable from chance, against 0.570 (0.554–0.585) for fusion. The intervals do not overlap. The composition model is below chance at 0.460 (0.447–0.474). After the dominant confound is matched away, a position-blind sequence model cannot separate real meQTLs from matched null pairs and this model can. That is the concrete content of the positional-resolution claim.

Figure discrimination_vs_distance.png
not present in this build.

Figure 5: Discrimination survives distance matching. AUROC for separating genome-wide significant meQTLs from tested-but-null pairs, unmatched and within distance strata, against the distance-only baseline.

Published architectures reach the same external signal, and that is the finding. ^{NEW} CpGenie and DeepCpG were scored on the identical held-out variant–CpG pairs through the identical evaluator and compared to fusion on a single shared distance-matched cohort, with paired block-bootstrap intervals on the difference (Section 2.5). Of twelve comparisons across the two cohorts and three metrics, one interval excludes zero: in GENOA, fusion exceeds DeepCpG on distance-matched AUROC by +0.018 (+0.004, +0.032). Fusion against CpGenie is +0.004 (−0.014, +0.020); signed ρ and direction agreement separate no pair of models in either cohort; and in the tissue-matched eGTEx cohort, where 418 significant pairs give intervals about ± 0.05 wide, every difference spans zero in both directions. This is convergence, not a failure to win. Three architectures designed independently over a decade recover the same external meQTL directions at the same strength, while a k -mer ridge on the same sequence sits at chance after matching. That the comparison returns chance for one comparator and agreement for three is what makes the agreement informative: the replicating signal belongs to sequence modelling rather than to this architecture, which speaks to whether the external validation reflects the cohorts or the model more directly than any margin would. It is also what the gating analysis predicts — the context tower is allele-invariant and cannot create a variant effect (Section 3.4), so the capacity separating fusion on absolute prediction is capacity the variant pathway does not use.

One asymmetry is worth recording. CpGenie is clearly behind DeepCpG on absolute prediction (beta MAE 0.1281 versus 0.1253, ROC-AUC 0.9406 versus 0.9437; Table 1) and is not behind it on variant effects. Accuracy on marginal methylation does not by itself establish fidelity to allele-specific sequence determinants, and the two capabilities should be evaluated separately rather than one being taken as evidence for the other.

The negative control behaves differently in the two cohorts, and we report both. ^{NEW} In GENOA the null stratum ($p >$

0.5, $n = 13,330$) gives $\rho = -0.004$ ($-0.021, +0.013$), covering zero. In eGTEx the corresponding stratum ($n = 20,964$) gives $\rho = +0.019$ ($+0.006, +0.032$), excluding it. The magnitude is negligible, and the eGTEx gradient decays monotonically ($0.348 \rightarrow 0.214 \rightarrow 0.109 \rightarrow 0.097 \rightarrow 0.033 \rightarrow 0.019$) rather than collapsing, which is what an underpowered cohort should produce: with 50–100 donors, $p > 0.5$ still contains many real but undetectable mQTLs, whereas GENOA’s $\sim 1,000$ donors make that stratum closer to genuinely null. We therefore rest the internal negative-control claim on GENOA and describe the eGTEx residual as sub-threshold signal, not as a second clean null.

What bounds the cross-tissue result. Two limits are worth stating precisely, because each has a mechanism and neither is a matter of sample size. First, direction agreement does not improve with the precision of the reference measurement: stratifying the significant pairs into quintiles of $|\hat{\beta}|/\text{SE}$ leaves agreement flat (0.521 in the least precisely measured quintile, 0.566 in the most, all intervals overlapping). The ceiling is therefore the model’s resolution, not noise in the reported effects, and cannot be recovered with better-measured meQTLs. Second, marginal discrimination does not exceed the distance baseline, as stated above; what survives matching is a smaller orthogonal component. The cross-tissue predictor is real, bounded, and we do not claim more for it than that.

3.4 Variant-effect prediction is sequence-driven, and therefore tissue-portable ^{NEW}

Gated fusion improves absolute methylation prediction substantially, but the variant-effect predictions above are carried entirely by the sequence tower. That is a structural property of the architecture rather than a shortcoming, and it has a useful consequence: because the variant pathway requires no tissue-specific input, it transfers to tissues for which no chromatin data exist — which is why the model can be scored against a blood cohort at all.

Fusion and sequence-only models were statistically equivalent on every variant-effect metric. Paired differences, bootstrapped over the same genomic blocks, were $+0.0026$ ($-0.0020, +0.0070$) for signed ρ , -0.0007 ($-0.0064, +0.0051$) for direction agreement, and -0.0021 ($-0.0053, +0.0011$) for within-distance-bin AUROC. The intervals are narrow enough to support equivalence rather than merely a failure to detect a difference: any difference is bounded within ± 0.007 .

This follows from the architecture. The context vector depends on the probe, not on the allele, and is therefore identical for wild-type and mutant input. Writing the fused embedding difference,

$$\Delta \mathbf{z} = [\tilde{\mathbf{h}}_{\text{seq}}^{\text{mut}} g_{\text{seq}}^{\text{mut}} - \tilde{\mathbf{h}}_{\text{seq}}^{\text{wt}} g_{\text{seq}}^{\text{wt}}] + \tilde{\mathbf{h}}_{\text{ctx}}[g_{\text{ctx}}^{\text{mut}} - g_{\text{ctx}}^{\text{wt}}],$$

the context term survives only through a gate shift that is itself driven by the sequence change. Context can modulate a variant effect; it cannot create one.

The gate is the only route by which allele-invariant context could still shape a prediction, and that route is also closed:

the fusion-minus-sequence difference is flat across quartiles of gate DNA share, with all eight intervals covering zero and the largest point estimate 0.007. A faint sign pattern runs in the predicted direction across quartiles but is not statistically supported.

This is consistent with the earlier observation that fusion and sequence rankings of the 440 synonymous candidates were highly concordant (signed $\rho = 0.981$, 96.6% sign agreement). The present analysis explains why, and bounds the effect.

Two practical consequences follow. Variant scoring requires only reference sequence, so it extends to any tissue or cell type without acquiring matched chromatin tracks — the property that made the cross-tissue GENOA evaluation possible. And the result generalises beyond this model: in any architecture that fuses sequence with allele-invariant covariates, the covariate branch cannot contribute to a paired wild-type/mutant contrast except through a gate shift, so reporting a fusion model’s variant-effect performance without its sequence-only ablation risks attributing to fusion what the sequence tower alone achieves.

3.5 Uncertainty is quantifiable, but the standard metric is confounded ^{NEW}

Because the framework is intended to prioritise variants, its predictions need an accompanying reliability estimate. Forward/reverse-complement disagreement — available at every locus at no additional cost, since both orientations are already scored — tracked per-locus error at Spearman $\rho \approx 0.50$ for the sequence and fusion models, and retains 75–83% of a three-seed ensemble’s incremental information from a single model.

Validating that claim exposed a trap worth reporting. On selective prediction the zero-parameter heuristic $|\hat{\beta} - 0.5|$ outperformed every genuine estimator, including the ensemble, in 8 of 9 model-seed combinations. This is a property of the metric, not of the estimators: absolute β error is mechanically compressed near 0 and 1, so a model merely confident at extreme predictions appears well calibrated. Repeating the comparison on the unbounded M scale separates the explanations — the heuristic’s within-stratum correlation collapses from $+0.0369$ to $+0.0099$, while cross-seed standard deviation ($+0.0915$ to $+0.0952$) and FWD–RC disagreement ($+0.0678$ to $+0.0793$) are unchanged (Fig. 6). **Scale-instability is the signature of a metric artifact:** a genuine uncertainty signal ranks error about equally well on either scale. Since β MAE is the field’s routine reporting choice, we assess calibration in logit space throughout and recommend the same for any bounded bimodal target.

3.6 A second evaluation domain: TCGA-BRCA tumours ^{NEW}

Because the model’s inputs depend on genomic position rather than disease state, its predictions are identical for normal and

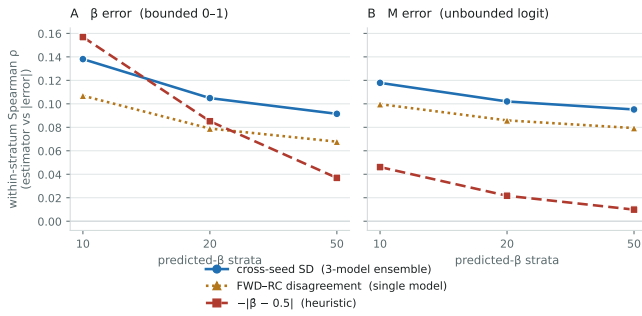


Figure 6: Scale stability separates uncertainty signal from metric artifact. Within-stratum rank correlation between each uncertainty estimator and absolute error, on the bounded β scale and the unbounded M scale. Genuine estimators are scale-stable; the boundary heuristic is not.

tumour tissue; scoring 699 unpaired TCGA-BRCA tumours therefore holds predictions fixed and varies only the target, measuring the component of the tumour methylome determined by sequence and reference chromatin context. Fusion β MAE moved from 0.0975 to 0.1180 and ROC-AUC from 0.9696 to 0.9448. Sixty-five per cent of held-out probes differ by less than 0.05 between the two targets, and degradation is confined to those that moved: stratifying on $|\beta_{\text{tumour}} - \beta_{\text{normal}}|$, error against the tumour target rises from 0.0769 to 0.4082 across strata while error against the normal target stays near 0.13, so the effect is target movement rather than probe difficulty. Most of the tumour methylome at these probes is thus recoverable without observing the tumour. The context-only model showed no degradation, which we report without explanation.

3.7 The model’s motif response is compositional, and a k -mer control shows why that matters ^{NEW}

Across 831 tested motifs, the strongest couplings between motif score change and predicted methylation change form a coherent group, all with the same sign (Table S4, Fig. 7). Negative coupling means motif weakening accompanied by predicted *hypermethylation*, the direction expected for factors whose binding protects CpGs from methylation.

The group is narrower than a family. Every one of the top eleven matrices contains the ETS core GGAA or its reverse complement TTCC, including the two zinc-finger factors ZBTB2 (ACCGAAGTG) and ZBTB11 (CACTCCGG), whose presence reflects that core rather than independent corroboration from a different structural class. Top-15 mean pairwise Jaccard overlap of covered variant sets is 0.278. The effective number of independent findings is one: a single 4-bp word.

A composition-only control identifies what this measures. ^{NEW} Repeating the analysis on the k -mer ridge baseline — a linear model with no attention, no pretraining and no representation of a binding site — reproduces the same signed couplings *more strongly* than the neural models (ELF4 -0.467

versus -0.266 ; FEV -0.466 versus -0.257 ; GABPA -0.377 versus -0.233). Standardised against each model’s own null distribution the two are equivalent. The coupling therefore measures sensitivity to GGAA sequence content, not learned binding-site grammar, and we describe it as such.

Two further results support that reading. Disruption *magnitude* predicts nothing (continuous $\rho = -0.011$, -0.023 to $+0.002$), where a graded binding-site mechanism predicts that breaking a site more should shift the prediction more. And meQTL discrimination is not concentrated in strongly disrupted sites (AUROC 0.592 versus 0.596).

The same control separates what the neural model uniquely contributes. ^{NEW}

On meQTL discrimination the k -mer baseline falls to 0.516 (0.496–0.535) and 0.509 (0.486–0.533), close to chance, against 0.592 and 0.596 for fusion. The two capabilities dissociate: the baseline is *better* at motif coupling and far worse at separating real meQTLs from null pairs. A bag-of- k -mers representation is position-blind — substituting a base changes the same counts wherever it falls in the window — and so cannot express the distance dependence reported in Table S3, which is the dominant structure in the data. What the pretrained model contributes to variant-effect prediction is positional resolution rather than sequence composition.

This has a methodological consequence beyond this model. A PWM disruption-coupling analysis cannot by itself demonstrate that a sequence model has learned regulatory grammar, because a linear composition model reproduces the result. Such analyses need a composition-only control.

The result survives three controls, each addressing a distinct alternative explanation.

It is not a global offset. Median coupling across all 831 motifs is -0.0037 (IQR -0.045 to $+0.038$), 51.4% of motifs are negative, and 185 are significantly negative against 155 significantly positive at $q < 0.05$. The null is centred and symmetric, placing the ETS group roughly five interquartile ranges outside it.

It is not GC or distance. Partial correlations are unchanged or slightly stronger. A library-wide gradient exists ($\rho = -0.20$ between motif GC and coupling), but the ETS motifs are mid-GC (0.44–0.65); were composition driving the result, GC-rich motifs at 0.85–0.90 would rank highest, and they do not.

It is not substitution bias. ETS cores are purine-rich, so weakening one typically implies a G/A→C/T change, and pyrimidine-rich sequence is more methylated on average. Residualising on canonical substitution class leaves every coupling essentially unchanged.

Two null results accompany the positive one and are reported with it. The *magnitude* of disruption predicts nothing (continuous $\rho = -0.011$, -0.023 to $+0.002$; strong versus weak median $|\Delta\hat{M}|$ 0.0177 versus 0.0184), and meQTL discrimination is not concentrated among strong disruptors (AUROC 0.592, 0.570–0.614, versus 0.596, 0.574–0.620). The model did not learn that disrupting motifs matters in general; it learned a signed, family-specific relationship.

Figure motif_disruption.png
not present in this build.

Figure 7: Motif disruption and predicted variant effects. Left: per-factor coupling between change in relative motif score and $\Delta\hat{M}$, with 95% block-bootstrap intervals; asterisks mark factors with previously reported methylation-sensitive binding. Right: meQTL discrimination among strongly versus weakly disrupting variants, which does not differ.

Two constraints on interpretation. First, the top-ranked factors are not independent: mean pairwise Jaccard overlap of their covered-variant sets is 0.278 and the maximum is 0.792, so this is the ETS core motif represented by several matrices rather than several independent findings. Second, the 18 factors with previously reported methylation-sensitive binding have median rank 136 of 831 – enriched relative to a chance expectation of 416, but decisively outranked by ETS.

3.8 From ranked variants to testable biological hypotheses

The preceding sections establish what the predictions are worth in aggregate. The purpose of the framework, however, is to act on individual variants: to take a specific substitution in a disease-relevant gene and produce a concrete, falsifiable statement about local methylation that an experiment can test. This section shows that end-to-end path, for a synonymous variant and a nonsynonymous one. Split status is stated for each, because it determines what the prediction is worth: the *NCOA2* candidate lies on a held-out chr8–9 probe, whereas both *STK11* probes fall in the training split. The *STK11* predictions are consequently hypothesis generation at loci the model has already seen, not held-out evidence, and no effect-size claim rests on them.

Applied as a prioritization engine to 440 synonymous variant–CpG pairs, the fusion model produced consistent rankings across seeds ($r = 0.681$ – 0.699). Median $|\Delta\hat{\beta}|$ was 0.00360 at promoter/TSS variants, 0.00261 at UTRs, and 0.00165 in gene bodies; variants within 50 bp of the target CpG had median 0.00512 against 0.00111 at 251–500 bp (Fig. 8). Neither genomic region nor distance was supplied as a feature.

The highest-ranked synonymous candidate, chr8:g.70148394G>A in *NCOA2*, is a GGC>GGT substitution 9 bp from probe cg20699548, which lies in an OpenSea context in the top quartile of both ATAC-seq and H3K27ac

Figure candidate_response_by_context.png
not present in this build.

Figure 8: Candidate response by biological context. Fusion difference magnitude grouped by (A) variant region and (B) distance to CpG.

signal. The ensemble predicted a hypomethylating shift of $\Delta\hat{\beta} = -0.1798 \pm 0.0198$, exceeding all 66 matched synonymous background variants (Fig. S1). *NCOA2* encodes a nuclear receptor coactivator with reported involvement in breast tumour growth [29]; no published evidence links this specific variant to *NCOA2* activation, and the prediction constitutes a testable hypothesis rather than a finding.

The framework extends to any single-nucleotide change. Among studied nonsynonymous variants in breast-cancer genes, the two highest-ranked were both in *STK11* [30]: rs2145420809 ($\Delta\hat{\beta} = 0.0774$ at cg16601904, immediately adjacent) and rs148928808 ($\Delta\hat{\beta} = -0.0613$ at cg08681293, 204 bp downstream), the latter a known pathogenic truncating variant. Both target CpGs lie on training chromosomes, so this is an illustrative application rather than validation.

3.9 Translational scope: portability, calibrated confidence, and the limits of variant classification ^{NEW}

Three findings bear on whether this framework could be used rather than only reported, and one of them is a boundary rather than a capability.

Predictions transfer across ancestry. Methylation and variant-effect predictors are overwhelmingly developed and evaluated in European-ancestry data, and failure to transfer is a recognised barrier to equitable genomic medicine. Section 3.3 provides a direct measurement rather than an assumption: the pooled estimate across a European-ancestry breast cohort and an African American whole-blood cohort is $\rho = 0.178$ (0.068–0.283) with no detectable heterogeneity ($I^2 = 0\%$). Ancestry and tissue differ together between these cohorts and cannot be separated (Limitations), so the defensible statement is the practical one: performance does not degrade when the model is moved to an African American cohort.

Confidence is calibrated, on the appropriate scale. Section 3.5 shows that reverse-complement disagreement tracks error, and that calibration must be assessed on the logit scale

because a zero-parameter boundary heuristic outperforms genuine estimators on the bounded β scale. A predictor that cannot indicate when it is unreliable is not usable for individual variant interpretation, so this is a prerequisite rather than an accessory result.

Predicted effect magnitude does not classify clinically annotated variants. We tested this directly, twice, under pre-registration (Section 2.11). For 35 held-out non-truncating ClinVar-pathogenic variants, none exceeded its matched background at the 0.05 tail (0 of 35, $p = 0.42$); that cohort spans only six distinct probes, with 25 variants at a single CpG, so its effective sample size is six and the test is uninformative whatever background is used. For 32 breast-cancer GWAS risk variants on 32 distinct probes, 3 exceeded the tail against 1.6 expected ($p = 0.21$), and the probe-clustered mean matched-background percentile was 54.9 (44.8–65.2, $p = 0.36$). The second test was adequately controlled – matching held substitution class fixed for 25 of 32 variants, and no context feature differed materially between cohort and background (largest $|d| = 0.36$) – so it is a clean null rather than an ambiguous one. Its power was 0.50 at a true mean percentile of 60, so it constrains the effect to roughly ten percentile points and does not establish absence.

Taken together these delimit the intended use. The framework predicts and ranks the *methylation consequence* of a substitution, with calibrated uncertainty and demonstrated transfer across tissue and ancestry. It does not identify pathogenic or risk variants from predicted effect magnitude alone, and we report that boundary rather than leaving it to be discovered.

4 Discussion

SilentMethyl unifies two tasks in regulatory genomics: predicting baseline CpG methylation and estimating how that state responds to single-nucleotide variation. Integrating local sequence with breast epigenomic context consistently improved methylation prediction over either modality alone, with the largest gains at CpG shores and at loci with high accessibility or H3K27ac signal—regions where methylation is expected to reflect an interaction between sequence and chromatin state rather than sequence alone [1, 2, 31].

^{REV}The independent variant evaluation is the central addition of this revision, and it supports a specific and bounded claim. On 39,657 variants from a cohort differing in tissue, ancestry and platform generation, predicted effects recovered direction above chance and correlated with reported effects in signed rank, and discrimination of true meQTLs from tested-but-null pairs survived matching on variant–CpG distance. The effect sizes are modest and we do not present them otherwise. Their value lies in three properties the earlier 81-association analysis could not provide: the scale to stratify rather than pool, an internal negative control in the form of a null stratum that returns to chance, and intervals that respect linkage disequilibrium.

The distance analysis deserves emphasis because it is the most likely source of an inflated claim in this literature. Pre-

dicted effect magnitude is strongly related to variant–CpG proximity, true meQTLs are closer to their CpGs than null pairs, and distance alone therefore reaches AUROC 0.595—indistinguishable from the model’s unmatched estimate. Reporting the unmatched number in isolation would substantially overstate what the model contributes. We report it alongside the baseline and alongside the distance-matched estimate, which is the quantity that reflects the model’s own information.

^{REV}Two findings generalise past this model. The first is methodological: uncertainty evaluation on bounded bimodal targets is confounded by range compression, so that a zero-parameter heuristic can outperform genuine uncertainty estimators on the β scale while collapsing on the logit scale. Because β MAE is the standard reporting choice in methylation prediction, any model that is simply confident at extreme predictions will appear well calibrated under current practice. Scale stability provides a cheap diagnostic, and logit-space evaluation a straightforward remedy.

The second is architectural. Gated multimodal fusion improved absolute methylation prediction substantially but contributed nothing measurable to variant-effect prediction, with paired equivalence intervals bounded within ± 0.007 and no rescue through the gate. The explanation is structural rather than empirical: the context vector is identical for reference and alternate input, so it cannot create a variant effect. The implication for the field is concrete—adding richer or better-matched epigenomic context to a model of this form will not improve variant-effect prediction, whereas making context features themselves allele-dependent might. Notably, this also means that the mismatch between the breast context features used here and the blood tissue of the validation cohort is not what limits variant-effect performance.

^{REV}The motif analysis supplies the mechanistic layer the earlier draft lacked. A model given no motif information, trained only to predict methylation, recovered a signed and family-specific relationship: weakening an ETS-family binding site accompanies predicted hypermethylation. ETS and GABPA occupancy is an established hallmark of unmethylated CpG-island promoters, so the biology itself is not new. What is new is the route — unsupervised recovery from sequence alone, detectable on held-out probes in a different tissue, ancestry and platform generation — and the quantitative ordering, in which previously characterised methylation-sensitive factors are enriched relative to chance yet decisively outranked. We state the distinction explicitly because the alternative reading, that this constitutes a new regulatory discovery, would not survive scrutiny and is not what the data support.

The accompanying null results constrain the claim usefully. Disruption magnitude does not predict effect size, and meQTL discrimination is no higher among strong disruptors. A model that had learned a general sensitivity to motif damage would show both. That neither appears, while a signed family-level coupling does, is more informative than the positive result alone, and is the reason we report the three together.

Limitations

One boundary belongs first. This architecture is not established as predicting variant effects better than earlier ones: on paired intervals against CpGenie and DeepCpG over identical pairs, eleven of twelve comparisons are indistinguishable. What the external cohorts establish is that the meQTL signal is real and reproducible, not that it is best recovered here.

Methylation targets derive from cohort medians of 97 TCGA solid-tissue-normal breast samples collected adjacent to tumours, which may retain field effects or cell-composition heterogeneity [18, 32]. Epigenomic context comes from MCF-10A reference tracks rather than matched samples, and HM450 covers only a subset of genomic CpGs.

The external validation cohort is whole blood while the model’s context features are breast; results on GENOA are therefore cross-tissue transfer rather than tissue-matched validation. GENOA effect sizes are reported on a normalized-phenotype scale, so rank and sign comparisons are meaningful while magnitude calibration against them is not available.

Motif occurrences are predicted from position weight matrices, not measured binding; without matched ChIP-seq in the relevant tissue these remain candidate sites. The top-ranked factors share a core motif and are not independent observations (mean pairwise Jaccard overlap 0.278 of their covered-variant sets), so the finding concerns one motif family rather than a set of factors. The tumour evaluation holds model inputs fixed and varies only the target, and measures the sequence-determined component of the tumour methylome rather than generalisation to a new input domain.

The cross-ancestry claim is confounded with tissue. GENOA differs from the reference data in ancestry *and* in tissue, so the absence of heterogeneity ($I^2 = 0\%$) shows that performance survives both differences together without establishing which, if either, the model is robust to individually.

The two matched-background tests were powered to detect a mean percentile of about 60 in the GWAS cohort and nothing at all in the ClinVar cohort, whose 35 variants span six probes. Neither result establishes the absence of an association between predicted magnitude and clinical annotation; they exclude effects larger than roughly ten percentile points in the better-powered case.

Ancestry-stratified error on the training cohort is not adequately powered: open genetic ancestry calls for the 97 training normals yield 84 European, 4 African, and 1 South Asian participant. Cross-ancestry evidence therefore rests on the contrast between the African American GENOA cohort and European-dominant reference data, which also differ in tissue and so do not constitute a clean ancestry contrast.

Evaluation currently uses a single chromosome-holdout design; repeated chromosome-blocked splits, regulatory-enrichment analysis, multi-cohort evaluation on unpaired TCGA-BRCA tumours, and conformal prediction intervals are in progress and are listed in the draft-status section.

5 Data and Code Availability

Code, run instructions, and Supplementary Data are available at <https://github.com/ilikecheezits/SilentMethyl>. Primary sources are the GDC, eGTEx, GENOA, ENCODE, GENCODE, UCSC, and the HM450 annotation resource [19, 21, 24, 25, 28, 31, 33]. Harmonised GENOA variant–CpG pairs, per-seed variant scores, and the evaluation outputs underlying Tables 2 and S2 are included in the Supplementary Data.

SilentMethyl predicts CpG methylation and the methylation differences caused by single-nucleotide variation in unseen chromosome regions, and its variant predictions agree with reported effects in direction and rank across cohorts differing in tissue, ancestry and platform. The two accompanying findings — that calibration on bounded bimodal targets must be assessed in logit space, and that gated fusion cannot contribute to variant-effect prediction while its context features remain allele-invariant — apply to any model built this way.

References

references.bib was not available when this draft was built, so citation keys are shown in place of formatted references. Building alongside the project bibliography restores the full list. Keys marked † are new in this revision and must be added.

References

- [1] jones2012functions
- [2] schubeler2015function
- [3] angermueller2017deepcpg
- [4] zhou2015predicting
- [5] avsec2021effective
- [6] zeng2017cpgenie
- [7] zhou2022interact
- [8] li2026deepmethylation
- [9] liu2025methven
- [10] avsec2025alphagenome †
- [11] ying2024methylogpt †
- [12] mallet2021rc
- [13] sauna2011understanding
- [14] stergachis2013exonic
- [15] alvarez2025mlh1
- [16] blackburn2016tp53
- [17] jankowska2022adamts13
- [18] tcga2012breast
- [19] grossman2016gdc
- [20] du2010betamvalue
- [21] zhou2017infinium
- [22] chen2013crossreactive
- [23] zhou2024dnabert2
- [24] frankish2021encode
- [25] shang2023genoa †
- [26] gemma2012 †
- [27] castrocolin2013jaspar †
- [28] oliva2023mqt1
- [29] cai2019ncoa2

[30] lipsa2019stk11
[31] encode2012integrated

[32] teschendorff2016field
[33] kent2002ucsc

Figure top_candidate_matched_background.png
not present in this build.

Figure S1: Top candidate versus matched background. The absolute response of the *NCOA2*-associated candidate compared with 66 matched observed synonymous variants.

Supplementary Figures and Tables

Table S1: Chromosome split dataset. Methylated means median $\beta > 0.5$.

Split	Loci	Mean β	Methylated
Train	345,359	0.4948	52.25%
Validation	46,557	0.5035	53.19%
Test	26,570	0.5029	53.20%

Table S2: Agreement by association strength. Fusion, three-seed ensemble, non-CpG-altering held-out pairs.

Stratum	n	Signed ρ	Direction
$p < 5 \times 10^{-8}$	4,037	+0.152 [+0.117, +0.184]	0.553
0.05–0.5	15,891	+0.023 [+0.006, +0.040]	0.508
$p > 0.5$	13,330	−0.004 [−0.020, +0.014]	0.497

Table S3: Direction agreement by variant–CpG distance. Fusion, three-seed ensemble, genome-wide significant non-CpG-altering held-out GENOA pairs. Brackets are 95% block-bootstrap intervals.

Distance from target CpG	n	Direction agreement
0–50 bp	844	0.568 [0.534, 0.601]
50–100 bp	479	0.605 [0.559, 0.654]
100–200 bp	761	0.558 [0.524, 0.594]
200–300 bp	685	0.545 [0.506, 0.582]
300–400 bp	642	0.555 [0.514, 0.598]
400–501 bp	626	0.497 [0.459, 0.535]

Table S4: Strongest per-factor coupling between motif disruption and predicted methylation change. Spearman ρ between the change in relative motif score and $\Delta\hat{M}$, among held-out non-CpG-altering pairs covered by each motif. Successive columns add controls. All $q_{\text{BH}} < 10^{-12}$.

Factor	n	ρ	+GC/dist	+subst.	motif GC
ELF4	1,005	−0.266	−0.268	−0.281	0.55
FEV	2,068	−0.257	−0.259	−0.257	0.56
EHF	1,711	−0.251	−0.253	−0.252	0.51
ELF1	1,352	−0.242	−0.246	−0.233	0.54
ETV1	1,606	−0.237	−0.242	−0.240	0.49
GABPA	968	−0.233	−0.235	−0.230	0.50